SCHEME and ANSWER KEY for 25901917
MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (Honours)/MGU-BBA (Honours)/MGU-BCA (Honours) Special Examinations, July 2026 (for 2024 Admission Regular Examinations, October 2025)
Third SEMESTER
Discipline Specific Core Course (DSC) - MG3DSCPHY200 - PRINCIPLES OF MECHANICS
(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)
Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A
Short Answer Type Questions
Answer any 10 questions
Each question carries 2 marks

1) Under what condition is Newton's second law F =ma applicable?
Newton's second law in the form F=ma is applicable only when:
The mass of the body remains constant during its motion.
The motion is observed in an inertial (non-accelerating) frame of reference. (2 marks)

2) A particle of mass 2 kg initially at rest, is subjected to a force F= $2e^{2t}$ Determine v(t) at t = 2s.
Mass=2 kg
Force F(t)=$2e^{2t}$N
Initial velocity, v(0)=0 Using Newton's second law,
F= ma, a =F/m
a = (2 x $2e^{2t}$)/2 = e^{2t} m/s^2
dv/dt = e^{2t} , solving we get v(2s) = 26.8 m/s (Formula-1 mark, correct answer with units -1mark)

3) If the air resistance force acting on an object is F = −0.1 v^2, what is the force when v = 10 m/s
F = −0.1 v^2
when v=10 m/s, F = -0.1 x (10)2 = -0.1 x 100 = -10 N
The negative sign indicates that the force acts opposite to the direction of motion. (correct answer with units -2 marks)

4) What is the difference between gravitational field intensity and gravitational potential?
The **gravitational field intensity** at a point is the gravitational force acting per unit mass placed at that point.
The **gravitational potential** at a point is the work done per unit mass in bringing a test mass from infinity to that point in the gravitational field. (Both definitions – 2 marks)

5) What is Newton's law of gravitation? Write the formula.
Newton's law of gravitation states that every two bodies in the universe attract each other with a force that is directly proportional to the product of their masses and inversely proportional to the square of the distance between their centres.
F = Gm_1m_2/r^2 (Both law and formula – 2 marks)

6) How can we determine the magnitude and direction of force from the field lines representation of the gravitational field?

The **direction** of the gravitational force is along the tangent to the field line, always towards the attracting mass.

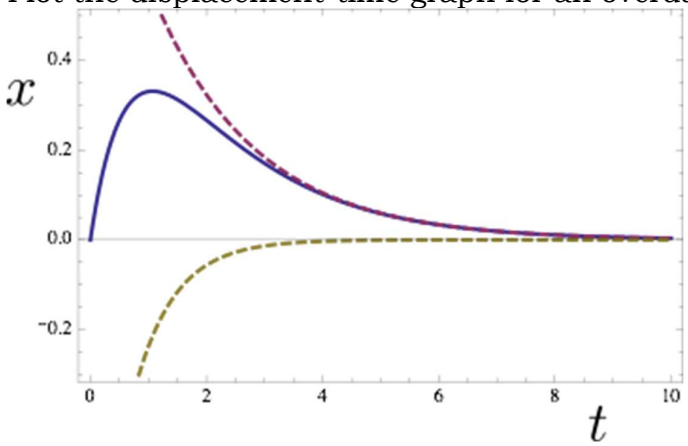
The **magnitude** of the gravitational force depends on the density of the field lines. Where the field lines are closer together, the force is stronger; where they are farther apart, the force is weaker. (Both points - 2 marks)

7) At what displacements is the potential energy of the simple harmonic oscillator is minimum and maximum? Explain.

Minimum at the equilibrium position and maximum at the extreme position (1 mark)

At the extreme positions, height is maximum, and velocity is zero; therefore, total energy is purely Potential energy. At the equilibrium position, velocity is maximum, and therefore total energy is purely kinetic. (1 mark)

8) Plot the displacement-time graph for an overdamped harmonic oscillator.



Total displacement is the sum of two terms, one exponentially increasing and another exponentially decreasing. Correct plot with contributing exponent term curves (2 marks)

Only the sum of curves (1 mark)

9) What is meant by a rigid body?

If all the points in a body maintain fixed relative distance from one another, then it is called a rigid body (2 marks)

10) Apply the concept of fictitious force to explain why passengers feel pushed outward in a turning car.

Passengers feel pushed outward because of inertia. When the car turns, a fictitious force called centrifugal force acts outwards. This causes the car to turn, but your body tends to continue the straight-line motion due to inertia. This leads to your body being pushed sideways.

11) Why does the day length on Earth slowly increase over millions of years?

Due to orbital boost, the rotational energy places the moon in higher orbits each year;s this causes a redistribution of mass and a momentum shift. This slows down the spin of the planet. (page 182; Patrik Hamill)

12) How can angular momentum of a system of particles be split into angular momentum of centre of mass and angular momentum about centre of mass.

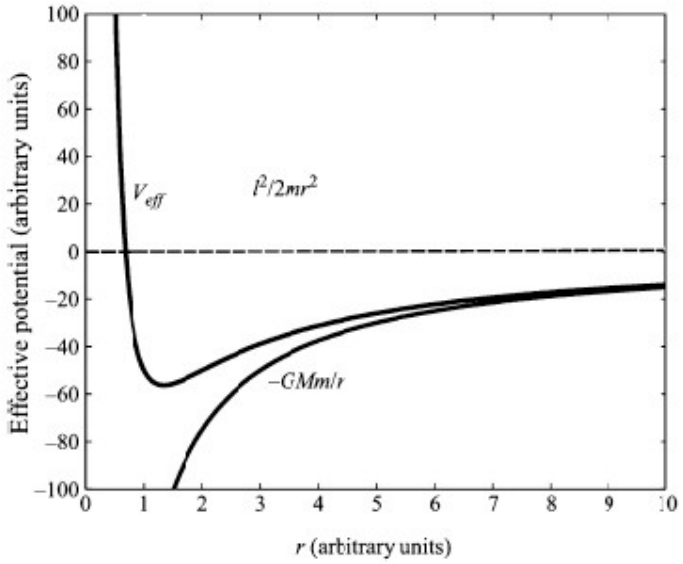
In a system of particles, the internal torques particles exert on each other do not affect the angular momentum of the system, just as internal forces do not affect linear momentum. Therefore, the total angular momentum can be considered as the angular momentum of the centre of mass and the angular momentum of these particles about the centre of mass.

13) How does Kepler's third law show that planets farther from the Sun take longer to revolve around it?

Kepler's third law states that the period of a planet squared is proportional to its semimajor axis cubed. Thus, farther planets have greater semimajor axes and longer time periods.

14) What is effective potential energy in central force motion? Illustrate it with an energy diagram for an attractive inverse square law field.

$$V_{eff} = \frac{l^2}{2mr^2} - \frac{GMm}{r}.$$



0.4 The effective potential energy $V_{eff}(r)$ is the sum of two terms, one positive and one negative.

Part B
Short Essay Type Questions
Answer any 6 questions
Each question carries 5 marks

15) Determine the motion of a particle if the force is a function of position of the particle

Page no: 64-65, Intermediate Dynamics by Patrick Hamil (Correct Answer – 5 marks, Partial answer - give marks accordingly)

16) Write a note on gravitational potential energy

Expand on the idea: The gravitational potential energy of an object of mass m at a distance r from the center of the Earth is $V = -GmM_e/r$, where M_e is the mass of the Earth. From this, obtain Newton's law of universal gravitation.

Gravitational potential energy – explanation (2 marks)
Equation with terms (1 mark)
Implications: 2 marks

17) Derive the time period of a simple pendulum, and explain why it is valid only for small oscillations.

Page no: 283-284, Intermediate Dynamics by Patrick Hamil (Correct derivation – 4 marks, explanation for small oscillation - 1 mark, Partial answer- give marks accordingly)

18) Write the differential equation and general solution of a damped harmonic

oscillator and discuss the case of an underdamped oscillator.

Page no: 291 - 294, Intermediate Dynamics by Patrick Hamil (Only setting the differential equation - 1 mark, obtaining up to general solution 3 marks, everything up to underdamped case – 5 marks, Partial answer -give marks accordingly)

19) State the differential equation of a forced damped oscillator and obtain the solution.

Page no: 297 - 305, Intermediate Dynamics by Patrick Hamil (setting the differential equation 2 marks, obtaining the solution 3 marks, partial credits for correct attempts)

20) What is meant by Coriolis force? A gun fires a bullet horizontally at speed v from a place on the North pole of Earth. In which direction will it deflect due to Coriolis force?

Explanation with equation: page no: 405 & 406 Intermediate Dynamics by Patrick Hamil, Definition of Coriolis force with equation- 2 marks.

The Coriolis force is given by

$F = -2m\ (\boldsymbol{\Omega} \times v)$, where $\boldsymbol{\Omega}$ is Earth's angular velocity and v is the bullet's velocity. In this case, since Earth's angular velocity is directed upwards and v is horizontal, Coriolis force is maximum. The bullet will be deflected to the right of its direction of motion due to the Coriolis force. (3 marks)

21) A particle of mass m has an initial velocity v_0 i. It is subjected to a constant It is subjected to a constant force $F = f_0$ j, If L is the angular momentum of the particle, then evaluate d^2L/dt^2 at the initial time.

Page No: 177-178, Intermediate Dynamics by Patrick Hamil

22) Obtain the equations of motion for a planet moving around the Sun in polar coordinates.

page no: 243- Intermediate Dynamics by Patrick Hamil

Obtaining the radial equation (1.5 marks)

Obtaining the Lagrangian (1.5 marks)

Final equation of motion (using Lagrangian) - 2 marks